Industriële Besturingen programmeren

Week 5 – Besturingsalgebra

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P2 - 2024/2025

DE HAAGSE HOGESCHOOL

Wetten van De Morgan

Geheugen + noodstop

Niet aangemeld Overleg Bijdragen Registreren Aanmelden

Artikel Overleg Lezen Bewerken Brontekst bewerken Geschiedenis Doorzoek Wikipedia

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Wetten van De Morgan

De **wetten van De Morgan**, of **regels van De Morgan**, zijn twee wetten in de formele logica die een verband leggen tussen de beide logische operatoren *EN* en *OF* en de negatie. Deze relatie wordt ook de **dualiteit van De Morgan** genoemd. Zij zijn genoemd naar de Britse wiskundige Augustus De Morgan, maar waren al eerder bekend.

Voor twee proposities *A* en *B* luiden de wetten:

niet (*A* en *B*) = (niet *A*) of (niet *B*)

niet (*A* of *B*) = (niet *A*) en (niet *B*)

In symbolen, waarbij *EN* door · wordt voorgesteld, *OF* door + en *NIET* door een overstreping, wordt dat:

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

$$Q = (b1 + q) \cdot (\overline{a0 + \overline{n}})$$

$$Q = (b1 + q) \cdot (\overline{a0} \cdot \overline{\overline{n}})$$

$$Q = (b1 + q) \cdot \overline{a0} \cdot n$$

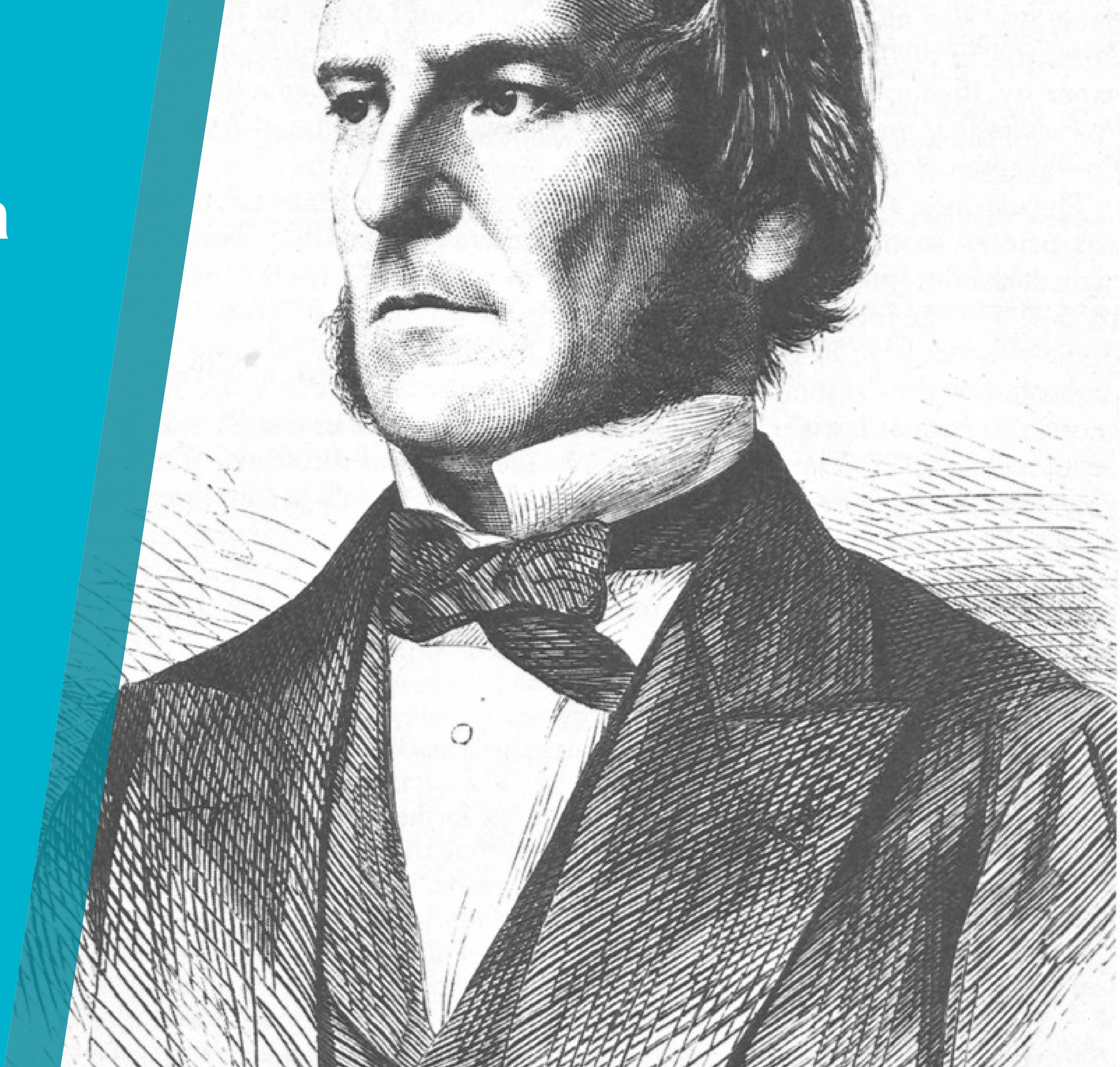
Besturingsalgebra

Talstelsels

Waarheidstabellen

Wetten en regels

Optimalisaties



Talstelsels

Hexadecimaal naar binair (en terug)

Voorbeeld: 4A

Spitsen: 4 A

Uitschrijven: 0100 1010

Antwoord: 01001010

Binair	Hex	Dec
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

Besturingsalgebra

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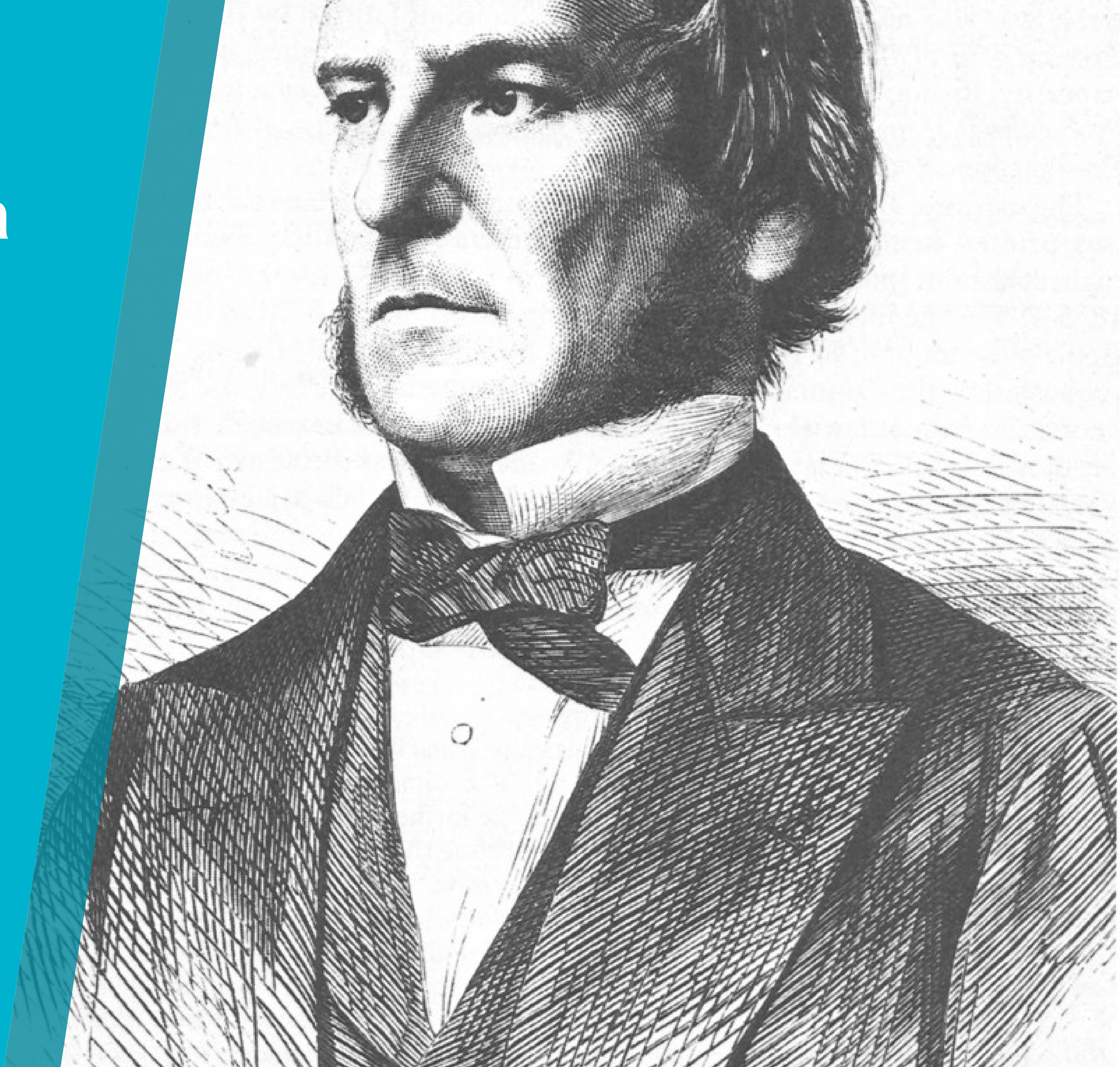


Waarheidstabellen

Overzicht en structuur

a	b	$X = a \cdot b$
0	0	0
0	1	0
1	0	0
1	1	1

Waarheidstabellen

Overzicht en structuur

Worden gebruikt voor:

1. Het bepalen of een (besturings)formule een ‘1’ aan de uitgang oplevert

2. Het bewijzen dat twee formules functioneel gelijk zijn

3. Tellen en de uitkomst van variabelencombinaties vergelijken

We bekijken de eerste twee toepassingen

Waarheidstabellen

Het bepalen of een (besturings)formule een ‘1’ aan de uitgang oplevert

c	b	a	$(\bar{a}+c)\cdot\bar{a}\cdot b$		X
0	0	0	$(1+0)\cdot 1\cdot 0$	$1\cdot 1\cdot 0$	0
0	0	1	$(0+0)\cdot 0\cdot 0$	$0\cdot 0\cdot 0$	0
0	1	0	$(1+0)\cdot 1\cdot 1$	$1\cdot 1\cdot 1$	1
0	1	1	$(0+0)\cdot 0\cdot 1$	$0\cdot 0\cdot 1$	0
1	0	0	$(1+1)\cdot 1\cdot 0$	$1\cdot 1\cdot 0$	0
1	0	1	$(0+1)\cdot 0\cdot 0$	$1\cdot 0\cdot 0$	0
1	1	0	$(1+1)\cdot 1\cdot 1$	$1\cdot 1\cdot 1$	1
1	1	1	$(0+1)\cdot 0\cdot 1$	$1\cdot 0\cdot 1$	0

Waarheidstabellen

Het bewijzen dat twee formules functioneel gelijk zijn

c	b	a	$(\overline{a} + c) \cdot \overline{a} \cdot b$	$\overline{a} \cdot b$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

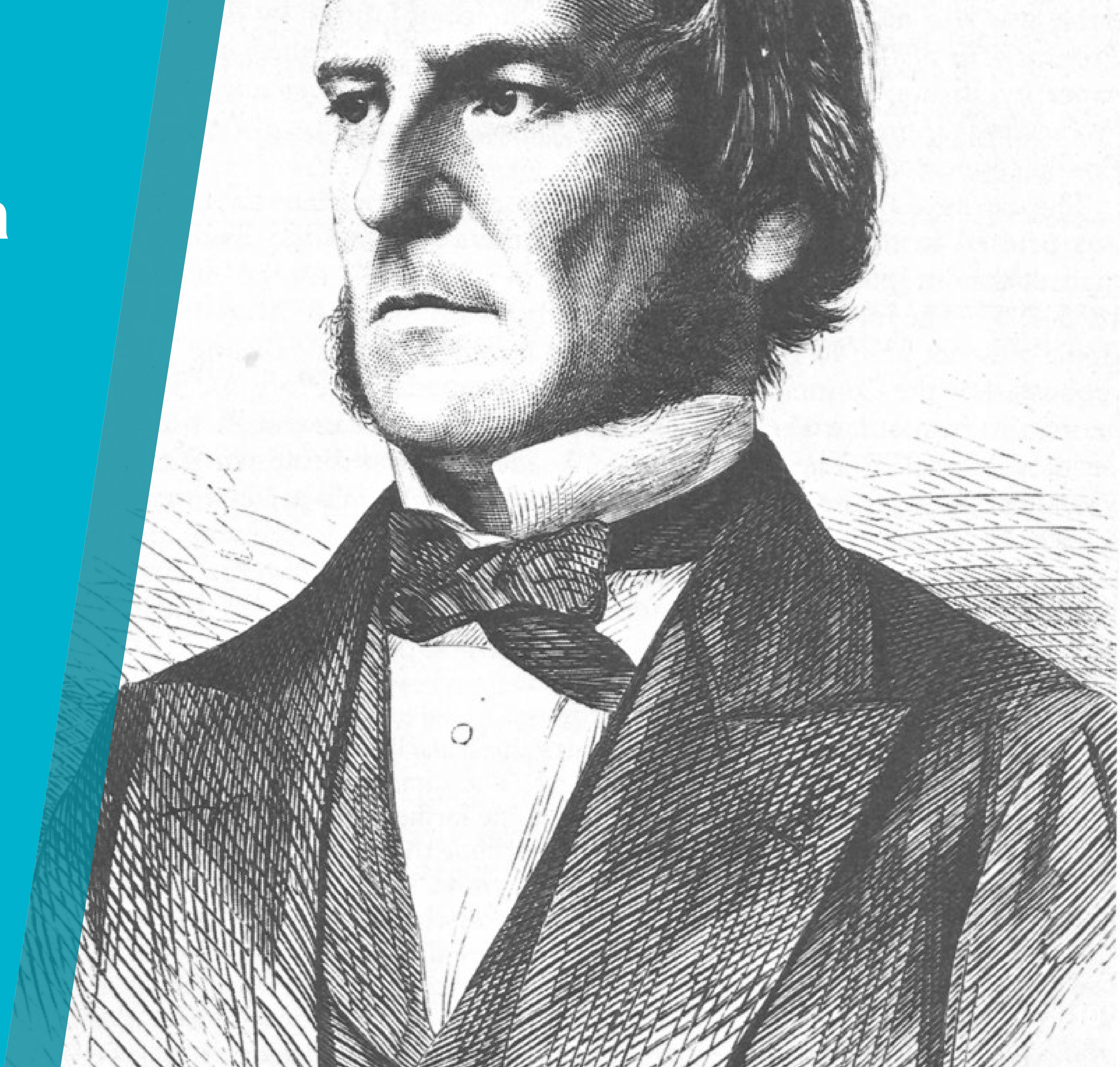
Besturingsalgebra

Talstelsels

Waarheidstabellen

Wetten en regels

Optimalisaties



Axioma's

Postulaten

Drie onbewezen grondstellingen van de besturingsalgebra:

Nummer	Axioma	Toelichting
1	$\overline{1} = 0$ $\overline{0} = 1$	NIET-functie
2	$0 \cdot 0 = 0$ $0 \cdot 1 = 0$ $1 \cdot 0 = 0$ $1 \cdot 1 = 1$	EN-functie
3	$0 + 0 = 0$ $0 + 1 = 1$ $1 + 0 = 1$ $1 + 1 = 1$	OF-functie

Wetten

Theorema's

Drie groepen met wetten van de besturingsalgebra:

Nummer	Wet	Toelichting
1	$a \cdot b = b \cdot a$	Commutatieve wetten
2	$a + b = b + a$	
3	$a \cdot (b \cdot c) = (a \cdot b) \cdot c$	Associatieve wetten
4	$a + (b + c) = (a + b) + c$	
5	$a \cdot b + a \cdot c = a \cdot (b + c)$	Distributieve wetten
6	$(a + b) \cdot (a + c) = a + b \cdot c$	

Wetten

Theorema's

Drie groepen met wetten van de besturingsalgebra: 1, 2, en 5 worden het meest toegepast.

Nummer	Wet	Toelichting
1	$a \cdot b = b \cdot a$	Commutatieve wetten
2	$a + b = b + a$	
3	$a \cdot (b \cdot c) = (a \cdot b) \cdot c$	Associatieve wetten
4	$a + (b + c) = (a + b) + c$	
5	$a \cdot b + a \cdot c = a \cdot (b + c)$	Distributieve wetten
6	$(a + b) \cdot (a + c) = a + b \cdot c$	

Regels

Afgeleid uit de axioma's en wetten (1/2)

Nummer	Regel	Naam
1	$\bar{a} \cdot a = 0$	Nul-elementen
2	$a \cdot 0 = 0$	
3	$\bar{a} + a = 1$	Eén-elementen
4	$a + 1 = 1$	
5	$a \cdot 1 = a$	Identiteiten
6	$a + 0 = a$	
7	$a \cdot a = a$	Herhalings- of gelijkheidsregels
8	$a + a = a$	

Regels

Afgeleid uit de axioma's en wetten (2/2)

Nummer	Regel	Naam
9	$\overline{a+b} = \bar{a} \cdot \bar{b}$	Morgan-regels
10	$\overline{a \cdot b} = \bar{a} + \bar{b}$	
11	$\overline{+} = \cdot$	Inversie-regels
12	$\overline{\cdot} = +$	
13	$\bar{\bar{a}} = a$	
14	$a + \bar{a} \cdot b = a + b$	Absorptie-regels
15	$\bar{a} + a \cdot b = \bar{a} + b$	
16	$a \cdot (\bar{a} + b) = a \cdot b$	
17	$\bar{a} \cdot (a + b) = \bar{a} \cdot b$	

Voorbeeld...

Wetten en regels toegepast

a. Vereenvoudig de volgende formule

b. Bewijs de gelijkwaardigheid

$$H = \bar{a} \cdot b \cdot c + a \cdot b \cdot c$$

Voorbeeld…

Wetten en regels toegepast

a. Vereenvoudig: $H = \bar{a} \cdot b \cdot c + a \cdot b \cdot c$

Stap	Uitkomst	Wet of regel
Originele formule	$H = \bar{a} \cdot b \cdot c + a \cdot b \cdot c$	
Wet 5:	$H = b \cdot c \cdot (\bar{a} + a)$	$a \cdot b + a \cdot c = a \cdot (b + c)$
Regel 3:	$H = b \cdot c \cdot 1$	$\bar{a} + a = 1$
Regel 5:	$H = b \cdot \mathrm{c}$	$a \cdot 1 = a$
Vereenvoudigde formule:	$H = b \cdot \mathrm{c}$	

Voorbeeld…

Wetten en regels toegepast

b. Bewijs de gelijkwaardigheid: $H = \bar{a} \cdot b \cdot c + a \cdot b \cdot c = b \cdot \mathrm{c}$

c	b	a	$\bar{a} \cdot b \cdot c + a \cdot b \cdot c$	H=	$b \cdot c$	H=
0	0	0	$1 \cdot 0 \cdot 0 + 0 \cdot 0 \cdot 0$	$0 + 0 = 0$	$0 \cdot 0$	0
0	0	1	$0 \cdot 0 \cdot 0 + 1 \cdot 0 \cdot 0$	$0 + 0 = 0$	$0 \cdot 0$	0
0	1	0	$1 \cdot 1 \cdot 0 + 0 \cdot 1 \cdot 0$	$0 + 0 = 0$	$1 \cdot 0$	0
0	1	1	$0 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 0$	$0 + 0 = 0$	$1 \cdot 0$	0
1	0	0	$1 \cdot 0 \cdot 1 + 0 \cdot 0 \cdot 1$	$0 + 0 = 0$	$0 \cdot 1$	0
1	0	1	$0 \cdot 0 \cdot 1 + 1 \cdot 0 \cdot 1$	$0 + 0 = 0$	$0 \cdot 1$	0
1	1	0	$1 \cdot 1 \cdot 1 + 0 \cdot 1 \cdot 1$	$1 + 0 = 1$	$1 \cdot 1$	1
1	1	1	$0 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 1$	$0 + 1 = 1$	$1 \cdot 1$	1

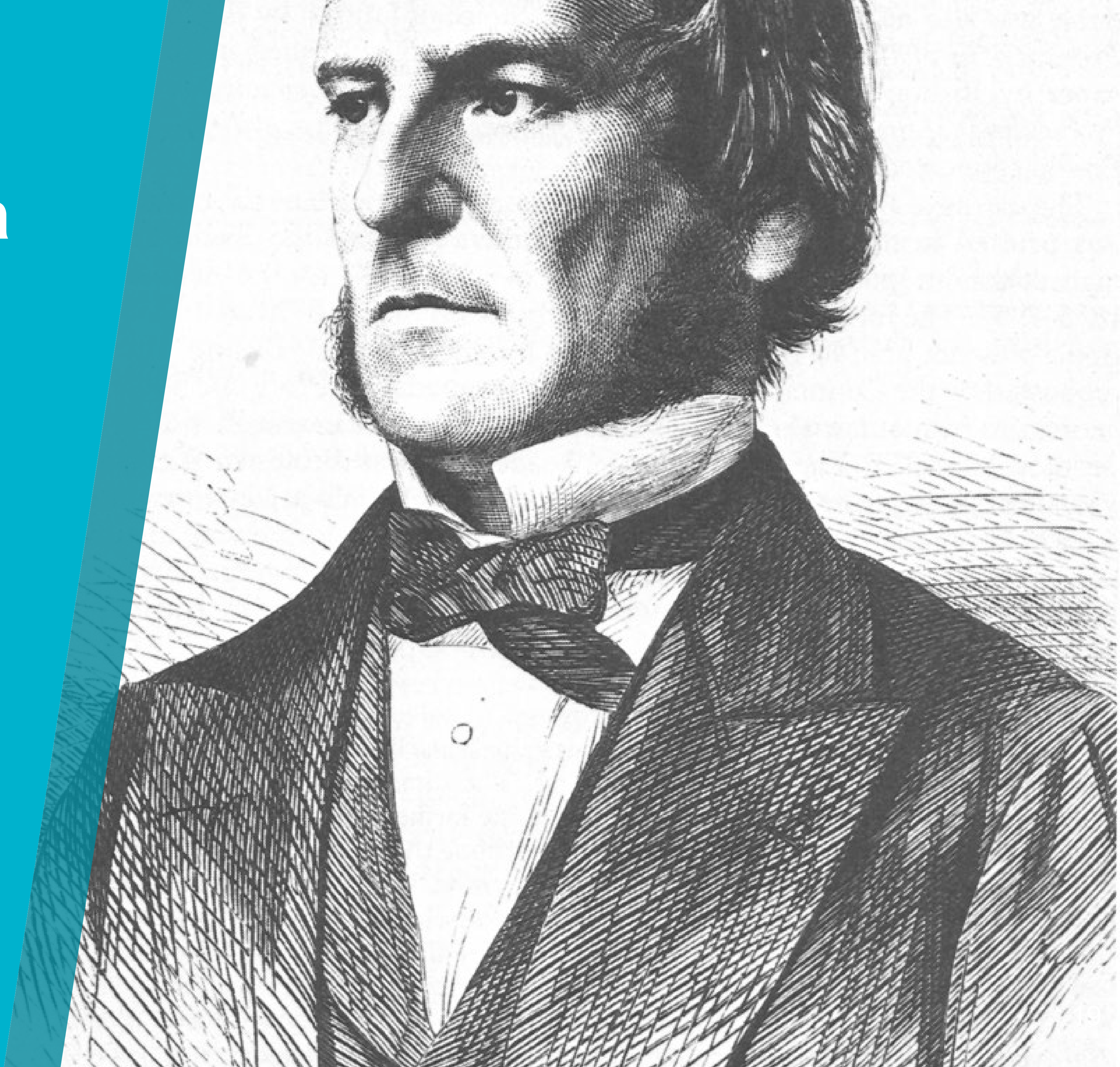
Besturingsalgebra

Talstelsels

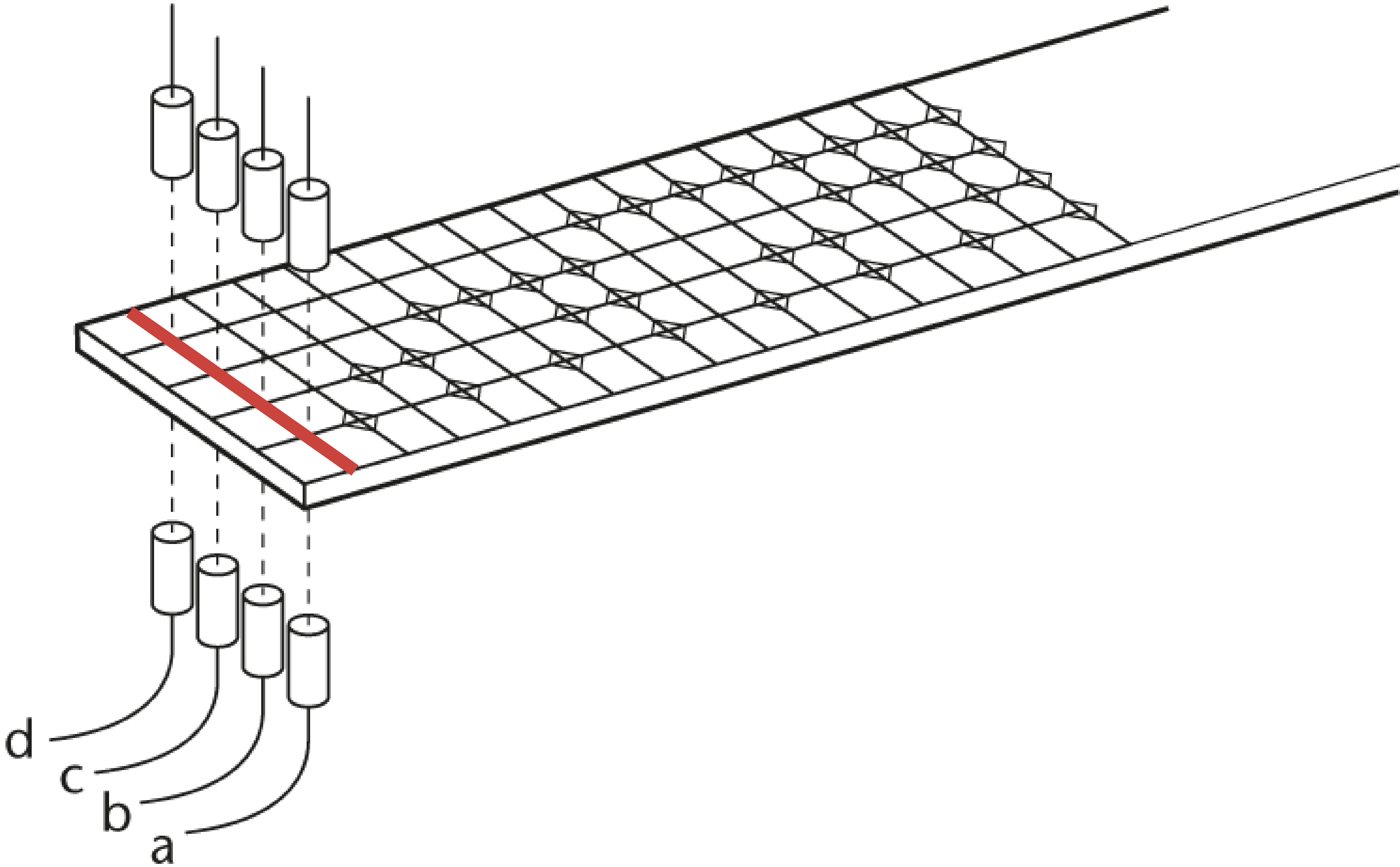
Waarheidstabellen

Wetten en regels

Optimalisaties

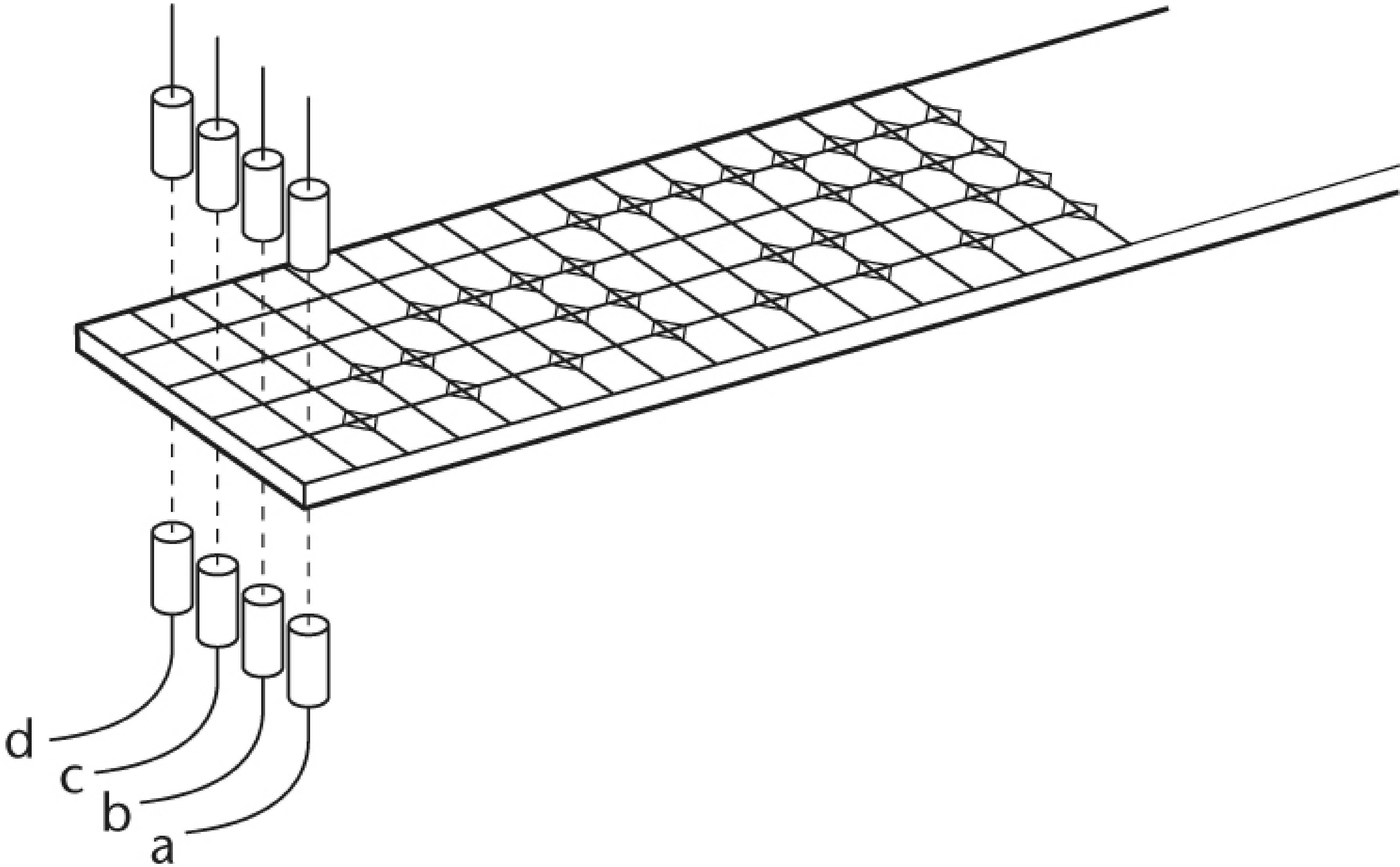


Meetliniaal



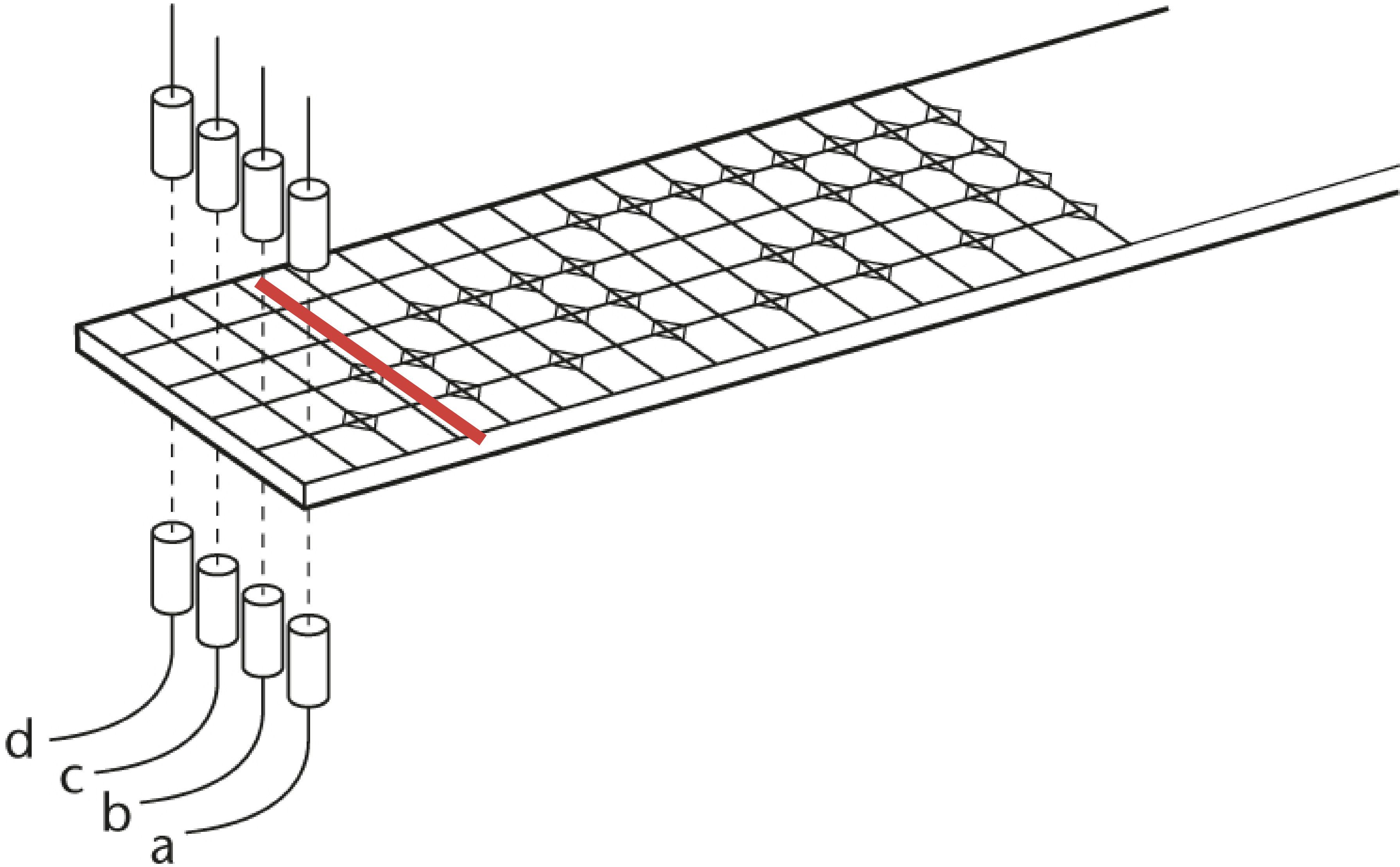
Meetliniaal

d	c	b	a
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
…	…	…	…



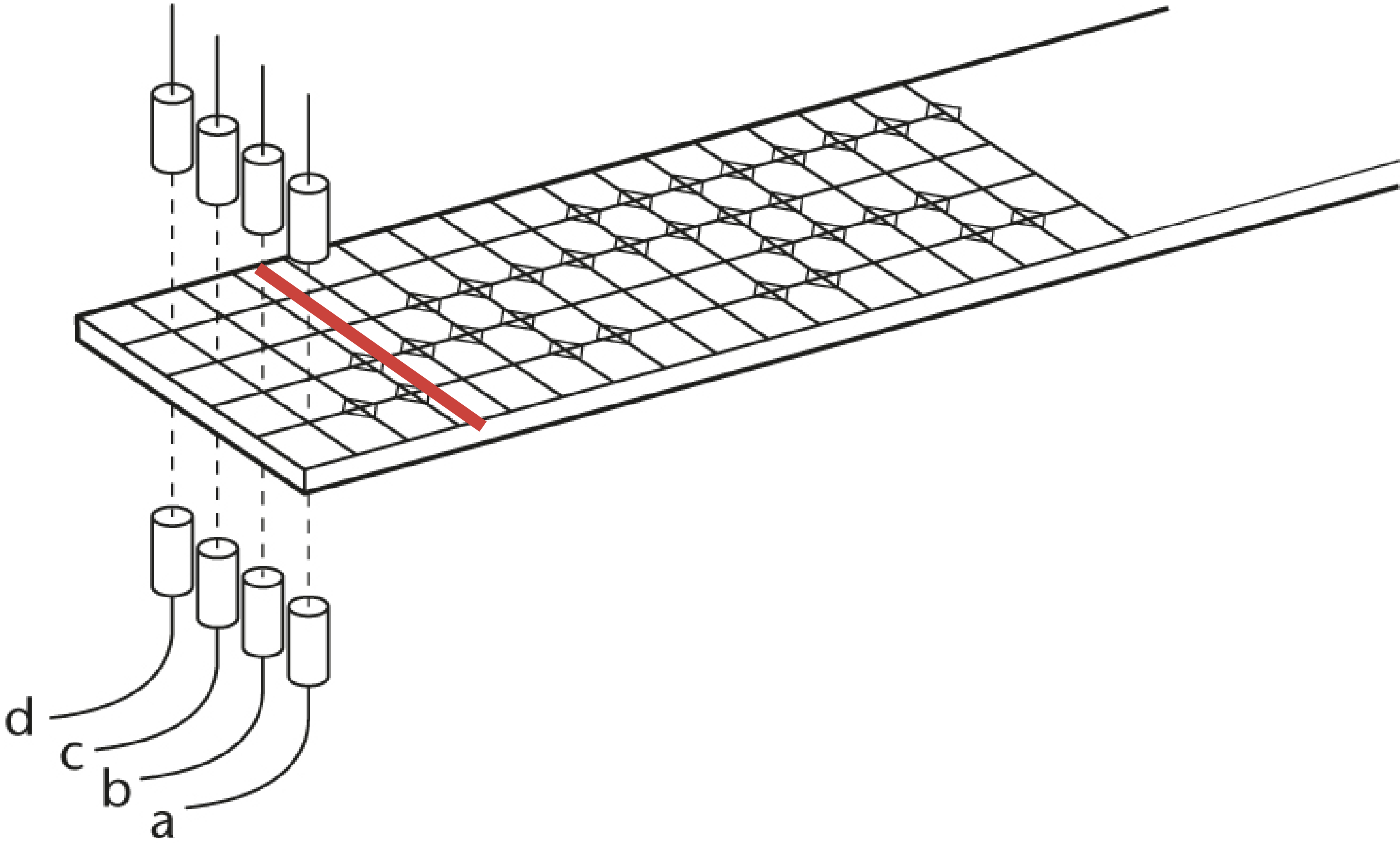
Meetliniaal

d	c	b	a
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
…	…	…	…



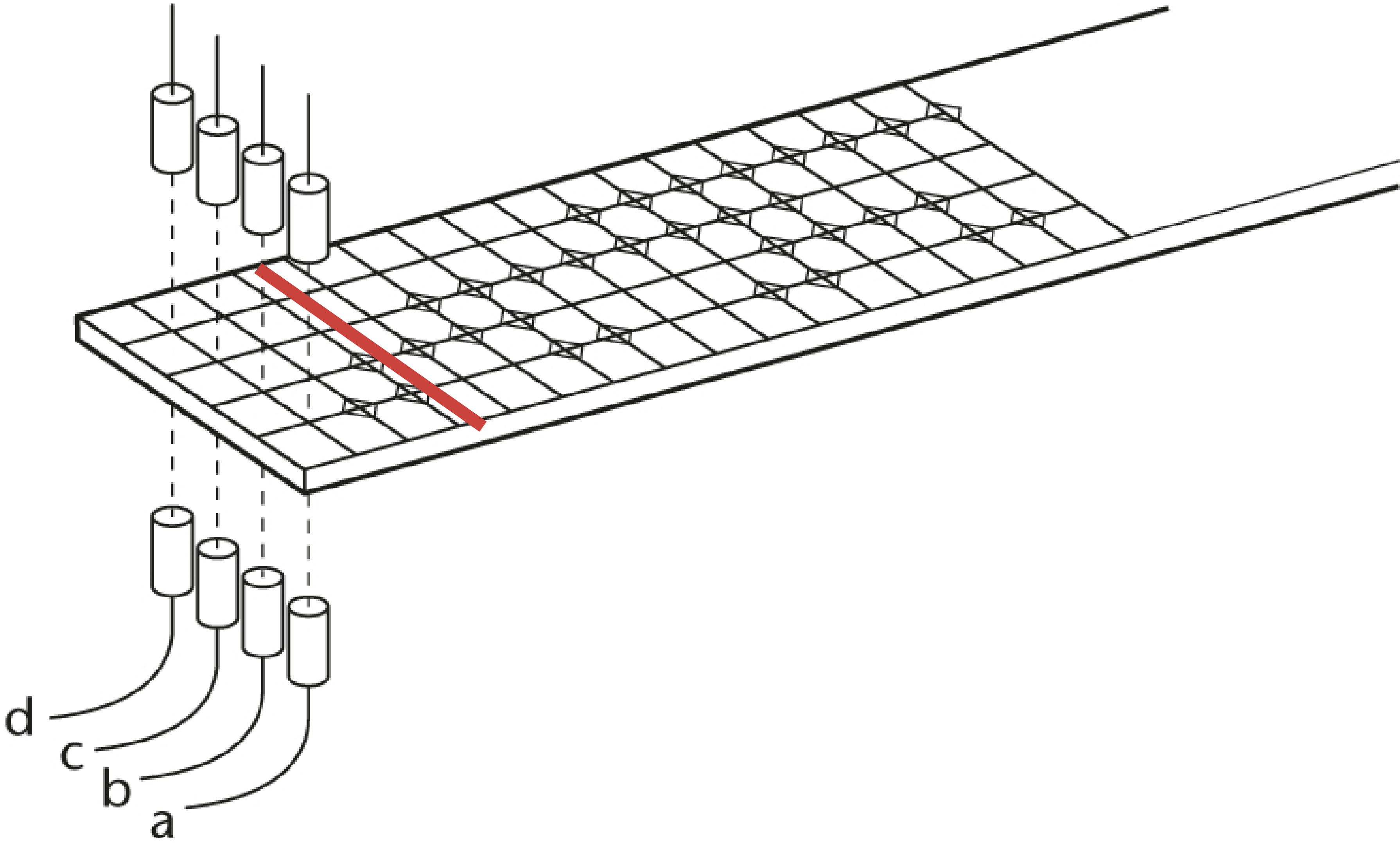
Meetliniaal

d	c	b	a
0	0	0	0
0	0	0	1
0	0	1	1
0	0	1	0
0	1	1	0
0	1	1	1
0	1	0	1
0	1	0	0
1	0	0	0
…	…	…	…



Meetliniaal

d	c	b	a
0	0	0	0
0	0	0	1
0	0	1	1
0	0	1	0
0	1	1	0
0	1	1	1
0	1	0	1
0	1	0	0
1	0	0	0
…	…	…	…



Gray-code

Spiegeltje spiegeltje aan de wand…

a
0
1

Gray-code

Spiegeltje spiegeltje aan de wand…

a
0
1

b	a
0	0
0	1
1	1
1	0

spiegellijn bit a

Gray-code

Spiegeltje spiegeltje aan de wand…

a
0
1

b	a
0	0
0	1
1	1
1	0

c	b	a	
0	0	0	
0	0	1	spiegellijn bit a
0	1	1	
0	1	0	spiegellijn bit a en b
1	1	0	
1	1	1	spiegellijn bit a
1	0	1	
1	0	0	

Gray-code

Spiegeltje spiegeltje aan de wand…

a
0
1

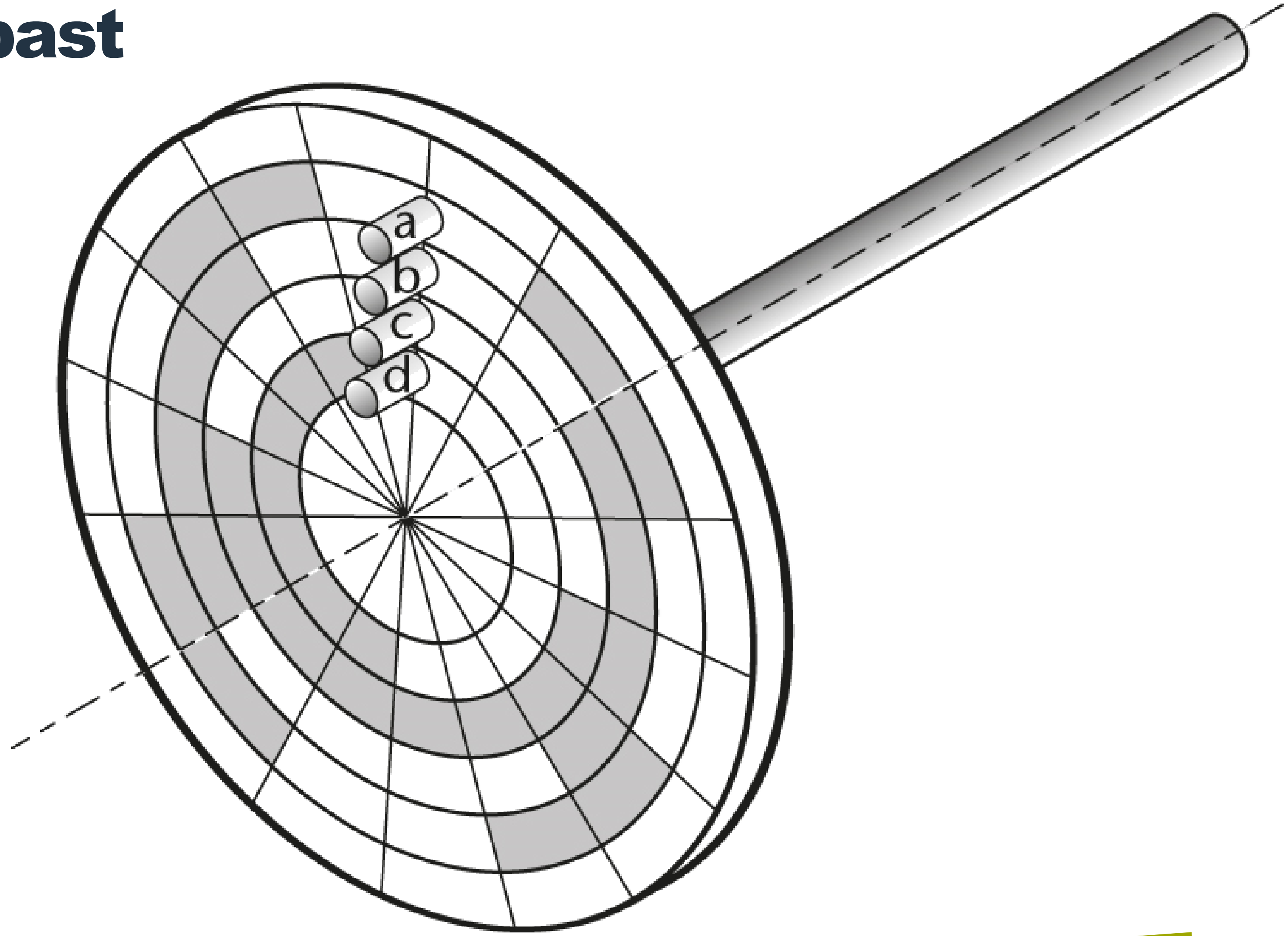
b	a
0	0
0	1
1	1
1	0

c	b	a
0	0	0
0	0	1
0	1	1
0	1	0
1	1	0
1	1	1
1	0	1
1	0	0

d	c	b	a	
0	0	0	0	
0	0	0	1	spiegellijn bit a
0	0	1	1	
0	0	1	0	spiegellijn bit a en b
0	1	1	0	
0	1	1	1	spiegellijn bit a
0	1	0	1	
0	1	0	0	spiegellijn bit a, b en c
1	1	0	0	
1	1	0	1	spiegellijn bit a
1	1	1	1	
1	1	1	0	spiegellijn bit a en b
1	0	1	0	
1	0	1	1	spiegellijn bit a
1	0	0	1	
1	0	0	0	

Gray-code toegepast

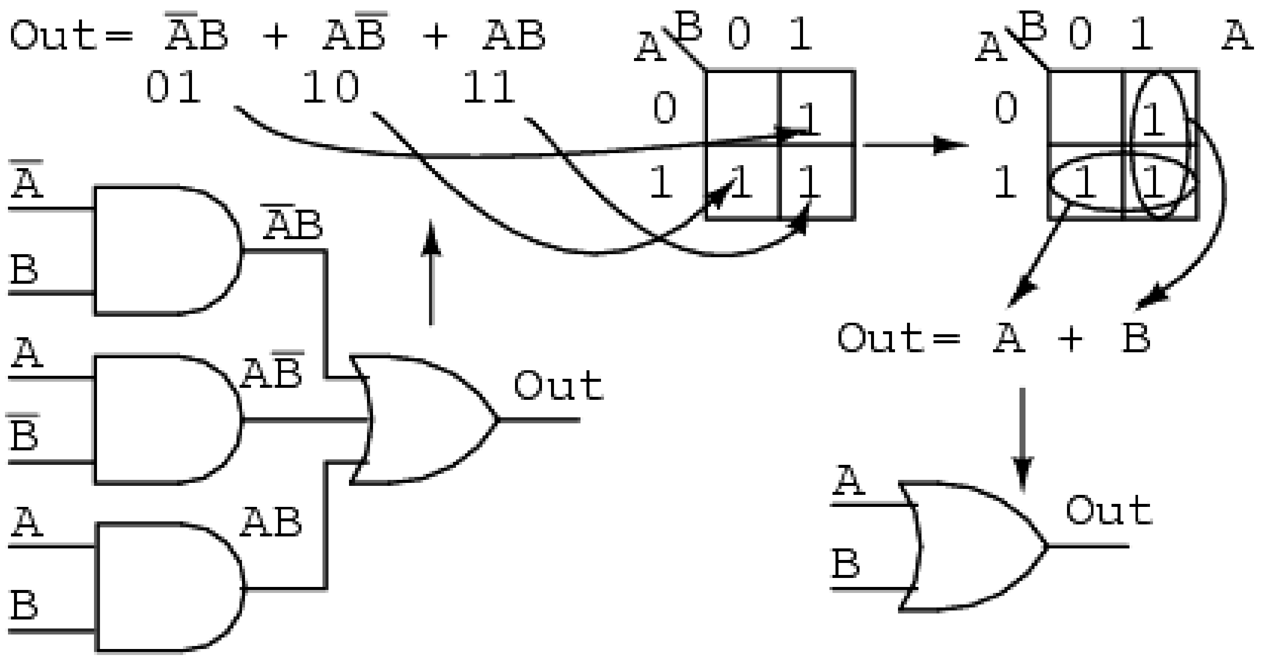
Digitale hoekmeting



Gray-code toegepast

Karnaugh-diagrammen

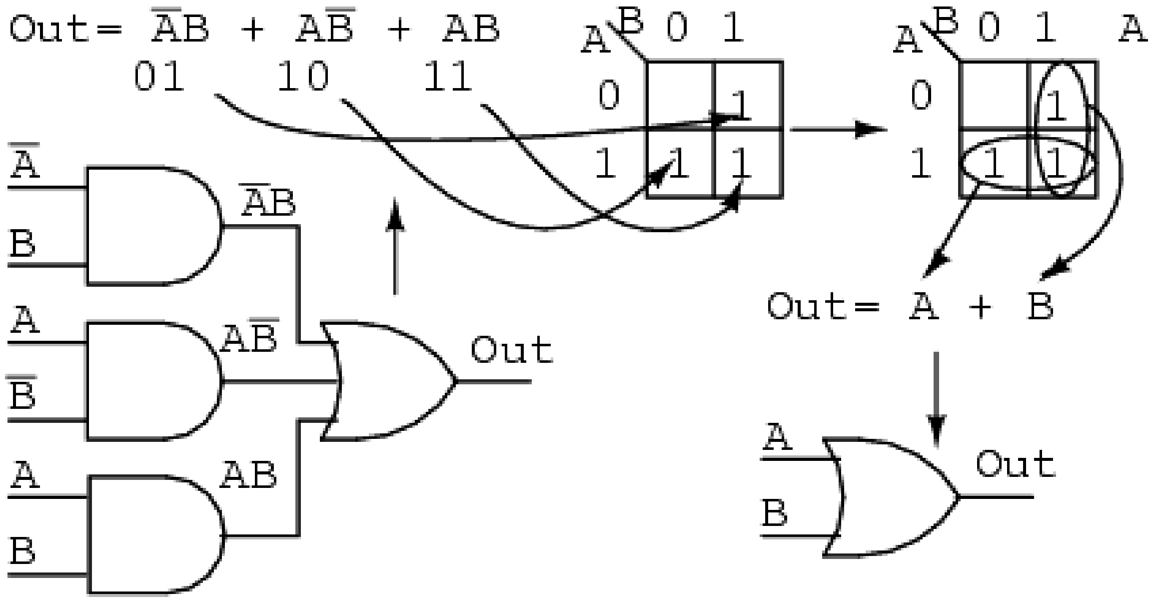
Karnaugh-diagrammen zijn ontwikkeld om besturingsformules snel en nauwkeurig te kunnen vereenvoudigen of samenstellen.



Gray-code toegepast

Karnaugh-diagrammen

b	a	$\bar{a} \cdot b + a \cdot \bar{b} + a \cdot b$	Out =	$a + b$	Out =
0	0	$1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0$	$0 + 0 + 0 = 0$	$0 + 0$	0
0	1	$0 \cdot 0 + 1 \cdot 1 + 1 \cdot 0$	$0 + 1 + 0 = 1$	$1 + 0$	1
1	1	$0 \cdot 1 + 1 \cdot 0 + 1 \cdot 1$	$0 + 0 + 1 = 1$	$1 + 1$	1
1	0	$1 \cdot 1 + 0 \cdot 0 + 0 \cdot 1$	$1 + 0 + 0 = 1$	$0 + 1$	1



Karnaugh-diagrammen

Minimaal- en maximaaltermen

Deze vereenvoudiging werkt enkel voor formules met minimaaltermen:

$$D = \bar{a} \cdot b \cdot c + a \cdot b \cdot \bar{c} + a \cdot \bar{b} \cdot c$$

Niet bij maximaaltermen:

$$D = (\bar{a} + b + c) \cdot (a + b + \bar{c}) \cdot (a + \bar{b} + c)$$

Wel kunnen maximaaltermen via de besturingsalgebra herleid worden tot miniaaltermen of eerst in een waarheidstabel worden weergegeven.

Karnaugh-diagrammen

Minimaal- en maximaaltermen

$$G = (a + \bar{b} + c) \cdot (b + \bar{c})$$

$$G = a \cdot b + a \cdot \bar{c} + \bar{b} \cdot b + \bar{b} \cdot \bar{c} + b \cdot c + c \cdot \bar{c}$$

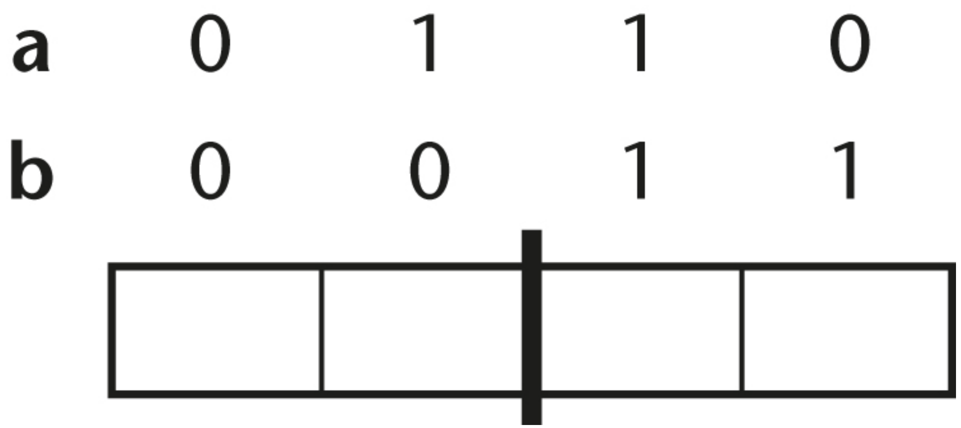
$$G = a \cdot b + a \cdot \bar{c} + 0 + \bar{b} \cdot \bar{c} + b \cdot c + 0$$

$$G = a \cdot b + a \cdot \bar{c} + \bar{b} \cdot \bar{c} + b \cdot c$$

Karnaugh-diagrammen

Van Gray naar Karnaugh

b	a
0	0
0	1
1	1
1	0



Karnaugh-diagrammen

...met twee variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
a)	$A = \overline{a} \cdot \overline{b} + a \cdot \overline{b}$	$\begin{array}{c\|c\|c\|c\|c} \mathbf{a} & 0 & 1 & 1 & 0 \\ \mathbf{b} & 0 & 0 & 1 & 1 \\ \hline & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{0} \end{array}$	$\begin{array}{c\|c\|c\|c\|c} \mathbf{a} & 0 & 1 & 1 & 0 \\ \mathbf{b} & 0 & 0 & 1 & 1 \\ \hline & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{0} \end{array}$	$A = \overline{b}$

Karnaugh-diagrammen

...met twee variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
a)	$A = \overline{a} \cdot \overline{b} + a \cdot \overline{b}$	a: 0 1 1 0 b: 0 0 1 1 **1 1 0 0**	a: 0 1 1 0 b: 0 0 1 1 **1 1 0 0**	$A = \overline{b}$
b)	$B = a \cdot \overline{b} + a \cdot b$	a: 0 1 1 0 b: 0 0 1 1 **0 1 1 0**	a: 0 1 1 0 b: 0 0 1 1 **0 1 1 0**	$B = a$

Karnaugh-diagrammen

...met twee variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
a)	$A = \bar{a} \cdot \bar{b} + a \cdot \bar{b}$	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 0 0**	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 0 0**	$A = \bar{b}$
b)	$B = a \cdot \bar{b} + a \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1 **0 1 1 0**	**a** 0 1 1 0 **b** 0 0 1 1 **0 1 1 0**	$B = a$
c)	$C = \bar{a} \cdot \bar{b} + a \cdot \bar{b} + a \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 1 0**	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 1 0**	$C =$ $a + \bar{b}$

Karnaugh-diagrammen

...met twee variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
a)	$A = \bar{a} \cdot \bar{b} + a \cdot \bar{b}$	**a** 0 1 1 0 **b** 0 0 1 1 **1** **1** **0** **0**	**a** 0 1 1 0 **b** 0 0 1 1 **1** **1** **0** **0**	$A = \bar{b}$
b)	$B = a \cdot \bar{b} + a \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1 **0** **1** **1** **0**	**a** 0 1 1 0 **b** 0 0 1 1 **0** **1** **1** **0**	$B = a$
c)	$C = \bar{a} \cdot \bar{b} + a \cdot \bar{b} + a \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1 **1** **1** **1** **0**	**a** 0 1 1 0 **b** 0 0 1 1 **1** **1** **1** **0**	$C =$ $a + \bar{b}$
d)	~~D = a · b … + a · b~~	**a** 0 1 1 0 **b** 0 0 1 1 **1** **1** **1** **1**	**a** 0 1 1 0 **b** 0 0 1 1 **1** **1** **1** **1**	$D = 1$

$$D = \bar{a} \cdot \bar{b} + a \cdot \bar{b} + a \cdot b + \bar{a} \cdot b$$

Karnaugh-diagrammen

...met twee variabelen

$$D = \bar{a} \cdot \bar{b} + a \cdot \bar{b} + a \cdot b + \bar{a} \cdot b$$

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
a)	$A = \bar{a} \cdot \bar{b} + a \cdot \bar{b}$	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 0 0**	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 0 0**	$A = \bar{b}$
b)	$B = a \cdot \bar{b} + a \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1 **0 1 1 0**	**a** 0 1 1 0 **b** 0 0 1 1 **0 1 1 0**	$B = a$
c)	$C = \bar{a} \cdot \bar{b} + a \cdot \bar{b} + a \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 1 0**	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 1 0**	$C =$ $a + \bar{b}$
d)	~~D = ā · b ... + a · b̄~~	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 1 1**	**a** 0 1 1 0 **b** 0 0 1 1 **1 1 1 1**	$D = 1$
e)	$E = \bar{a} \cdot \bar{b} + \bar{a} \cdot b$	**a** 0 1 1 0 **b** 0 0 1 1	**a** 0 1 1 0 **b** 0 0 1 1	$E = ???$

Karnaugh-diagrammen

Voorwaarden bij het vormen van groepen

Groepen vormen:

- Een groep bestaat enkel uit enen (geen nullen)
- Een groep enen bestaat uit 2^n termen (1, 2, 4, etc..)
- Een groep moet zo groot mogelijk gevormd worden
- De twee uiterste termen in een diagram liggen naast elkaar
- Een term mag in meerdere groepen voorkomen (groepen mogen overlappen)

Karnaugh-diagrammen

Voorwaarden bij het uitlezen

Formule vormen:

- Een variabele die binnen een groep van waarde veranderd, vervalt
- Een variabele die in alle velden van een groep als ‘1’ voorkomt, wordt als een JA-functie gelezen.
- Een variabele die in alle velden van een groep als ‘0’ voorkomt, wordt als een NIET-functie gelezen
- De uitgelezen groepen worden met een OF-functie tot een formule samengevoegd

Karnaugh-diagrammen

...met drie variabelen

<table>
<tr><th></th><th>Stap 1</th><th>Stap 2</th><th>Stap 3</th><th>Stap 4</th></tr>
<tr><th></th><th>Formule opstellen</th><th>Formule intekenen</th><th>Groepen vormen</th><th>Uitlezen</th></tr>
<tr>
<td>f)</td>
<td>$F = \overline{a} \cdot \overline{b} \cdot \overline{c} + a \cdot \overline{b} \cdot \overline{c}$</td>
<td>
<table>
<tr><td>a</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>b</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>c</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
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</table>
</td>
<td>$F = \overline{b} \cdot \overline{c}$</td>
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</table>

Karnaugh-diagrammen

...met drie variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
f)	$F = \overline{a} \cdot \overline{b} \cdot \overline{c} + a \cdot \overline{b} \cdot \overline{c}$	a 0 1 1 0 b 0 0 1 1 c 0 1 1 0 0 1 0 0 0 0	a 0 1 1 0 b 0 0 1 1 c 0 1 1 0 0 1 0 0 0 0	$F = \overline{b} \cdot \overline{c}$
g)	$G = a \cdot \overline{b} \cdot \overline{c} + a \cdot \overline{b} \cdot c$	a 0 1 1 0 b 0 0 1 1 c 0 0 1 0 0 1 0 1 0 0	a 0 1 1 0 b 0 0 1 1 c 0 0 1 0 0 1 0 1 0 0	$G = a \cdot \overline{b}$

Karnaugh-diagrammen

...met drie variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
f)	$F = \bar{a} \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot \bar{c}$	$\begin{array}{ccccc} & a\ 0 & 1 & 1 & 0 \\ & b\ 0 & 0 & 1 & 1 \\ c & & & & \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{array}$	$\begin{array}{ccccc} & a\ 0 & 1 & 1 & 0 \\ & b\ 0 & 0 & 1 & 1 \\ c & & & & \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{array}$	$F = \bar{b} \cdot \bar{c}$
g)	$G = a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$	$\begin{array}{ccccc} & a\ 0 & 1 & 1 & 0 \\ & b\ 0 & 0 & 1 & 1 \\ c & & & & \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{array}$	$\begin{array}{ccccc} & a\ 0 & 1 & 1 & 0 \\ & b\ 0 & 0 & 1 & 1 \\ c & & & & \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{array}$	$G = a \cdot \bar{b}$
h)	$H = \bar{a} \cdot \bar{b} \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot c +$ $+ \bar{a} \cdot b \cdot \bar{c} + \bar{a} \cdot b \cdot c$	$\begin{array}{ccccc} & a\ 0 & 1 & 1 & 0 \\ & b\ 0 & 0 & 1 & 1 \\ c & & & & \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{array}$	$\begin{array}{ccccc} & a\ 0 & 1 & 1 & 0 \\ & b\ 0 & 0 & 1 & 1 \\ c & & & & \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{array}$	$H = \bar{a}$

Karnaugh-diagrammen

...met drie variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
f)	$F = \bar{a} \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot \bar{c}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{array}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{array}$	$F = \bar{b} \cdot \bar{c}$
g)	$G = a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{array}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{array}$	$G = a \cdot \bar{b}$
h)	$H = \bar{a} \cdot \bar{b} \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot c +$ $+ \bar{a} \cdot b \cdot \bar{c} + \bar{a} \cdot b \cdot c$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{array}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{array}$	$H = \bar{a}$
i)	$I = ???$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{array}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{c} & & & & \\ 0 & & & & \\ 1 & & & & \end{array}$	$I = ???$

Karnaugh-diagrammen

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<tr><td>h)</td><td>$H = \bar{a} \cdot \bar{b} \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot c +$
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</table>

Karnaugh-diagrammen

Oefening

Vereenvoudig de onderstaande formule m.b.v. een Karnaugh-diagram:

$$S = a \cdot \bar{b} \cdot \bar{c} + a \cdot b \cdot c + a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot \bar{c}$$

a	0	1	1	0
b	0	0	1	1
c				
0				
1				

a	0	1	1	0
b	0	0	1	1
c				
0				
1				

Karnaugh-diagrammen

Oefening

En bewijs de gelijkwaardigheid

c	b	a	$a \cdot \bar{b} \cdot \bar{c} + a \cdot b \cdot c + a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot \bar{c}$	S=		S=
0	0	0	$0 \cdot 1 \cdot 1 + 0 \cdot 0 \cdot 0 + 0 \cdot 0 \cdot 1 + 1 \cdot 1 \cdot 1$	1		1
0	0	1	$1 \cdot 1 \cdot 1 + 1 \cdot 0 \cdot 0 + 1 \cdot 0 \cdot 1 + 0 \cdot 1 \cdot 1$	1		1
0	1	1	$1 \cdot 0 \cdot 1 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 + 0 \cdot 0 \cdot 1$	1		1
0	1	0	$0 \cdot 0 \cdot 1 + 0 \cdot 1 \cdot 0 + 0 \cdot 1 \cdot 1 + 1 \cdot 0 \cdot 1$	0		0
1	1	0	$0 \cdot 0 \cdot 0 + 0 \cdot 1 \cdot 1 + 0 \cdot 1 \cdot 0 + 1 \cdot 0 \cdot 0$	0		0
1	1	1	$1 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 0 + 0 \cdot 0 \cdot 0$	1		1
1	0	1	$1 \cdot 1 \cdot 0 + 1 \cdot 0 \cdot 1 + 1 \cdot 0 \cdot 0 + 0 \cdot 1 \cdot 0$	0		0
1	0	0	$0 \cdot 1 \cdot 0 + 0 \cdot 0 \cdot 1 + 0 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 0$	0		0

Karnaugh-diagrammen

Oefening

Vereenvoudig de onderstaande formule m.b.v. een Karnaugh-diagram:

$$S = a \cdot \bar{b} \cdot \bar{c} + a \cdot b \cdot c + a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot \bar{c}$$

c \ a b	0 0	1 0	1 1	0 1
0	1	1	1	0
1	0	0	1	0

c \ a b	0 0	1 0	1 1	0 1
0	1	1	1	0
1	0	0	1	0

$$S = \bar{b} \cdot \bar{c} + a \cdot b$$

Karnaugh-diagrammen

Oefening

Vereenvoudig de onderstaande formule m.b.v. een Karnaugh-diagram:

$$S = a \cdot \bar{b} \cdot \bar{c} + a \cdot b \cdot c + a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot \bar{c}$$

$$S = \bar{b} \cdot \bar{c} + a \cdot b$$

c	b	a	$a \cdot \bar{b} \cdot \bar{c} + a \cdot b \cdot c + a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot \bar{c}$	S=	$\bar{b} \cdot \bar{c} + a \cdot b$	S=
0	0	0	$0 \cdot 1 \cdot 1 + 0 \cdot 0 \cdot 0 + 0 \cdot 0 \cdot 1 + 1 \cdot 1 \cdot 1$	1	$1 \cdot 1 + 0 \cdot 0$	1
0	0	1	$1 \cdot 1 \cdot 1 + 1 \cdot 0 \cdot 0 + 1 \cdot 0 \cdot 1 + 0 \cdot 1 \cdot 1$	1	$1 \cdot 1 + 1 \cdot 0$	1
0	1	1	$1 \cdot 0 \cdot 1 + 1 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 + 0 \cdot 0 \cdot 1$	1	$0 \cdot 1 + 1 \cdot 1$	1
0	1	0	$0 \cdot 0 \cdot 1 + 0 \cdot 1 \cdot 0 + 0 \cdot 1 \cdot 1 + 1 \cdot 0 \cdot 1$	0	$0 \cdot 1 + 0 \cdot 1$	0
1	1	0	$0 \cdot 0 \cdot 0 + 0 \cdot 1 \cdot 1 + 0 \cdot 1 \cdot 0 + 1 \cdot 0 \cdot 0$	0	$0 \cdot 0 + 0 \cdot 1$	0
1	1	1	$1 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 0 + 0 \cdot 0 \cdot 0$	1	$0 \cdot 0 + 1 \cdot 1$	1
1	0	1	$1 \cdot 1 \cdot 0 + 1 \cdot 0 \cdot 1 + 1 \cdot 0 \cdot 0 + 0 \cdot 1 \cdot 0$	0	$1 \cdot 0 + 1 \cdot 0$	0
1	0	0	$0 \cdot 1 \cdot 0 + 0 \cdot 0 \cdot 1 + 0 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 0$	0	$1 \cdot 0 + 0 \cdot 0$	0

Karnaugh-diagrammen

...met vier variabelen

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Karnaugh-diagrammen

...met vier variabelen

	Stap 1	Stap 2	Stap 3	Stap 4
	Formule opstellen	Formule intekenen	Groepen vormen	Uitlezen
k)	$K = +\bar{a}\cdot\bar{b}\cdot c\cdot\bar{d} + \bar{a}\cdot\bar{b}\cdot c\cdot d + \bar{a}\cdot b\cdot c\cdot\bar{d} + \bar{a}\cdot b\cdot c\cdot d$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{d\,c} & & & & \\ 0\,0 & 0 & 0 & 0 & 0 \\ 0\,1 & 1 & 0 & 0 & 1 \\ 1\,1 & 1 & 0 & 0 & 1 \\ 1\,0 & 0 & 0 & 0 & 0 \end{array}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{d\,c} & & & & \\ 0\,0 & 0 & 0 & 0 & 0 \\ 0\,1 & 1 & 0 & 0 & 1 \\ 1\,1 & 1 & 0 & 0 & 1 \\ 1\,0 & 0 & 0 & 0 & 0 \end{array}$	$K = \bar{a}\cdot c$
l)	$L = +a\cdot\bar{b}\cdot c\cdot\bar{d} + b\cdot c\cdot\bar{d} + \bar{a}\cdot b\cdot c\cdot d + a\cdot\bar{b}\cdot c\cdot d + a\cdot b\cdot c$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{d\,c} & & & & \\ 0\,0 & 0 & 0 & 0 & 0 \\ 0\,1 & 0 & 1 & 1 & 1 \\ 1\,1 & 0 & 1 & 1 & 1 \\ 1\,0 & 0 & 0 & 0 & 0 \end{array}$	$\begin{array}{ccccc} & \mathbf{a}\ 0 & 1 & 1 & 0 \\ & \mathbf{b}\ 0 & 0 & 1 & 1 \\ \mathbf{d\,c} & & & & \\ 0\,0 & 0 & 0 & 0 & 0 \\ 0\,1 & 0 & 1 & 1 & 1 \\ 1\,1 & 0 & 1 & 1 & 1 \\ 1\,0 & 0 & 0 & 0 & 0 \end{array}$	$L = a\cdot c + b\cdot c = c\cdot(a+b)$

Karnaugh-diagrammen

...met vier variabelen

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<td>$K = \bar{a}\cdot c$</td></tr>
<tr><td>l)</td>
<td>$L = +a\cdot\bar{b}\cdot c\cdot\bar{d} + b\cdot c\cdot\bar{d} + \bar{a}\cdot b\cdot c\cdot d + a\cdot\bar{b}\cdot c\cdot d + a\cdot b\cdot c$</td>
<td><table>
<tr><td>a</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>b</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>d c</td><td></td><td></td><td></td><td></td></tr>
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<tr><td>0 1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1 1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
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<tr><td>d c</td><td></td><td></td><td></td><td></td></tr>
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<tr><td>0 1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
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</table></td>
<td>$L = a\cdot c + b\cdot c = c\cdot(a + b)$</td></tr>
<tr><td>m)</td>
<td>$M = +\bar{a}\cdot\bar{b}\cdot\bar{c}\cdot\bar{d} + \bar{a}\cdot\bar{b}\cdot c\cdot\bar{d} + \bar{a}\cdot\bar{b}\cdot\bar{c}\cdot d + \bar{a}\cdot b\cdot\bar{c}\cdot\bar{d} + \bar{a}\cdot b\cdot c\cdot\bar{d} + \bar{a}\cdot b\cdot\bar{c}\cdot d$</td>
<td><table>
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</table></td>
<td>$M = \bar{a}\cdot d + \bar{a}\cdot\bar{c} = \bar{a}\cdot(d + \bar{c})$</td></tr>
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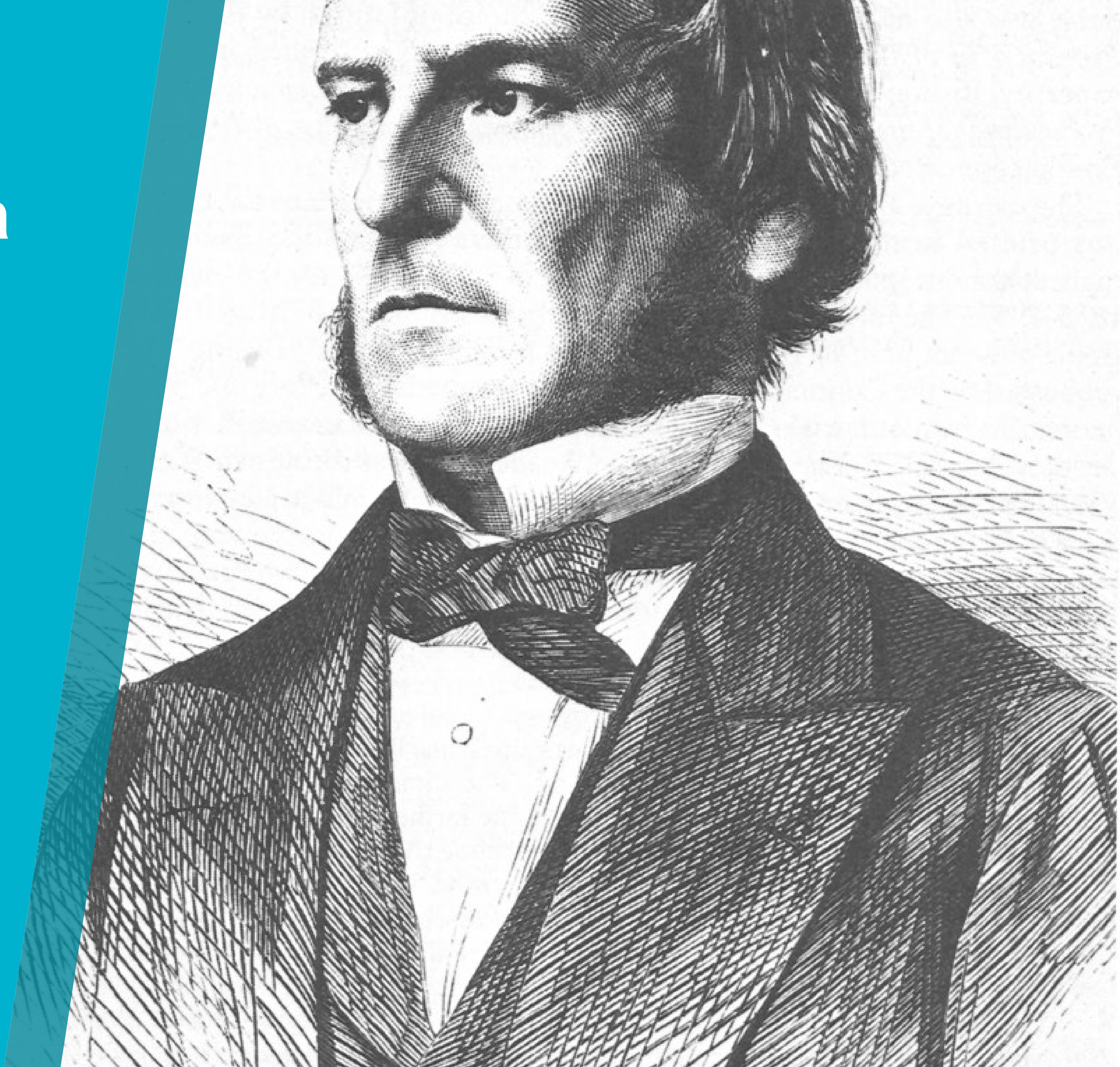
Besturingsalgebra



Talstelsels

Waarheidstabellen

Wetten en regels

Optimalisaties